

Homework Set HW13 due 4/1/04 at 11:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)

This is problem 6 Section 5.4 page 308.

Evaluate the definite integral

$$\int_{-1}^1 (x^3 + bx^2) dx = \underline{\hspace{2cm}}$$

2.(1 pt)

This is problem 8 Section 5.4 page 308.

Evaluate the definite integral

$$\int_{-2}^{-1} \frac{p}{x^2} dx = \underline{\hspace{2cm}}$$

3.(1 pt)

This is problem 12 Section 5.4 page 308.

Evaluate the definite integral

$$\int_0^1 (7x^8 + \sqrt{\pi}) dx = \underline{\hspace{2cm}}$$

4.(1 pt)

This is problem 16 Section 5.4 page 308.

Evaluate the definite integral

$$\int_{e^{-2}}^e \frac{dx}{x} = \underline{\hspace{2cm}}$$

5.(1 pt)

This is problem 18 Section 5.4 page 308.

Evaluate the definite integral

$$\int_0^1 \sqrt{t}(t - \sqrt{t}) dt = \underline{\hspace{2cm}}$$

6.(1 pt)

This is problem 20 Section 5.4 page 308.

Evaluate the definite integral

$$\int_1^4 \frac{x^2 + x - 1}{\sqrt{x}} dx = \underline{\hspace{2cm}}$$

7.(1 pt)

This is problem 24 Section 5.4 page 308.

Evaluate the definite integral

$$\int_0^{\pi/4} (\sec^2(x) - \tan^2(x)) dx = \underline{\hspace{2cm}}$$

8.(1 pt)

This is problem 26 Section 5.4 page 308.

Evaluate the definite integral

$$\int_1^2 \frac{x^3 + 1}{x} dx = \underline{\hspace{2cm}}$$

9.(1 pt)

This is problem 32 Section 5.4 page 308.

Find the area of the region under the curve over the prescribed interval

$$y = \sqrt{t} \text{ on } [0, 1]$$

AREA: $\underline{\hspace{2cm}}$ **10.(1 pt)**

This is problem 34 Section 5.4 page 308.

Find the area of the region under the curve over the prescribed interval

$$y = \sin(x) + \cos(x) \text{ on } [0, \pi/2]$$

AREA: $\underline{\hspace{2cm}}$ **11.(1 pt)**

This is problem 38 Section 5.4 page 308.

Find the area of the region under the curve over the prescribed interval

$$y = \frac{2}{1+t^2} \text{ on } [0, 1]$$

AREA: $\underline{\hspace{2cm}}$ **12.(1 pt)**

This is problem 40 Section 5.4 page 308.

Find the derivative of the given function

$$F(x) = \int_{-2}^x (t+1)\sqrt[3]{t} dt$$

$$F'(x) = \underline{\hspace{2cm}}$$

13.(1 pt)

This is problem 42 Section 5.4 page 308.

Find the derivative of the given function

$$F(t) = \int_t^2 \frac{e^x}{x} dx$$

$$F'(x) = \underline{\hspace{2cm}}$$

14.(1 pt)

This is problem 1 Section 5.5 page 315.

Evaluate the definite integrals

$$(a) \int_0^4 (2t+4) dt = \underline{\hspace{2cm}}$$

$$(b) \int_0^4 (2t+4)^{-1/2} dt = \underline{\hspace{2cm}}$$

15.(1 pt)

This is problem 6 Section 5.5 page 315.

Evaluate the indefinite integrals

$$(a) \int x(3x^2 - 5) dx = \underline{\hspace{2cm}} + C$$

$$(b) \int x(3x^2 - 5)^{50} dx = \underline{\hspace{2cm}} + C$$

16.(1 pt)

This is problem 8 Section 5.5 page 315.

Evaluate the indefinite integrals

$$(a) \int \frac{dx}{\sqrt{1-x^2}} = \underline{\hspace{2cm}} + C$$

$$(b) \int \frac{xdx}{\sqrt{1-x^2}} = \underline{\hspace{2cm}} + C$$

17.(1 pt)

This is problem 10 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \sqrt{3t-5} dt = \underline{\hspace{2cm}} + C$$

18.(1 pt)

This is problem 14 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \csc^2(5t) dt = \underline{\hspace{2cm}} + C$$

19.(1 pt)

This is problem 18 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \frac{(6x-9)}{(x^2-3x+5)^3} dx = \underline{\hspace{2cm}} + C$$

20.(1 pt)

This is problem 20 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \sin^3(t) \cos(t) dt = \underline{\hspace{2cm}} + C$$

21.(1 pt)

This is problem 24 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \frac{4x dx}{2x+1} = \underline{\hspace{2cm}} + C$$

22.(1 pt)

This is problem 30 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \frac{\ln(x+1)}{(x+1)} dx = \underline{\hspace{2cm}} + C$$

23.(1 pt)

This is problem 34 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$$\int \frac{e^{\sqrt{t}}}{\sqrt{t}(e^{\sqrt{t}}+1)} dt = \underline{\hspace{2cm}} + C$$

24.(1 pt)

This is problem 36 Section 5.5 page 315.

Evaluate the definite integral

$$\int_1^4 \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx = \underline{\hspace{2cm}} + C$$

25.(1 pt)

This is problem 40 Section 5.5 page 315.

Evaluate the definite integral

$$\int_0^2 x\sqrt{2x+1} dx = \underline{\hspace{2cm}} + C$$

26.(1 pt)

This is problem 21 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$$\frac{dy}{dx} = 3xy$$

NOTE: if your answer is expressed in implicit form as $y = Cf(x)$ where C is a constant, then your answer would be $Cf(x)$, e.g., you might have $C \sin(x^2)$ and in this case you would only enter $C \sin(x^2)$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$$y = \underline{\hspace{2cm}}$$

where C is a constant.

27.(1 pt)

This is problem 22 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$$\frac{dy}{dx} = \sqrt{y}x$$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y) - C$, e.g., you might have $\sin(x^2 - y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2 - y^2) - C$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$$f(x,y) - C = \underline{\hspace{2cm}}$$

where C is a constant.

28.(1 pt)

This is problem 24 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$$\frac{dy}{dx} = (y-4)^2$$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y) - C$, e.g., you might have $\sin(x^2 - y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2 - y^2) - C$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$$f(x,y) - C = \underline{\hspace{2cm}}$$

where C is a constant.

29.(1 pt)

This is problem 26 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$$\frac{dy}{dx} = \frac{y}{\sqrt{1-2x^2}}$$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y) - C$, e.g., you might have $\sin(x^2 - y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2 - y^2) - C$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$$f(x,y) - C = \underline{\hspace{2cm}}$$

where C is a constant.

30.(1 pt)

This is problem 30 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$$\frac{dy}{dx} = e^{y-x}$$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y) - C$, e.g., you might have $\sin(x^2 - y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2 - y^2) - C$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$$f(x,y) - C = \underline{\hspace{2cm}}$$

where C is a constant.

